先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 03 – Generative Classifiers and Bayesian Decisions

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The alarm is on. Is the part defective?

A teaching scenario

A factory inspects 10,000 parts. Only 2% are defective.

The sensor

It alarms on 90% of defective parts and 5% of sound parts.

Your first guess

After an alarm: 90%, 50%, or less than 50% defective?

What we will build in this lecture

Question	Tool we need
What does an alarm tell us?	Bayesian decision criterion
Which mistake matters more?	Minimum-risk decision
How can several clues be combined?	Naive Bayes classifier
Where do probabilities come from?	Parameter estimation
How do continuous features form a class?	Gaussian classification
How do we fit the class shapes?	Mean and covariance estimation

Finish line. Fit a classifier, explain its boundary, and choose an action.

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Separate the state, the observation, and the action

Object	In the factory	Notation
State	Defective or sound	Y
Observation	Alarm or no alarm	X
Action	Release, quarantine, or review	a

One important distinction

An alarm is observed. A defect is still unknown. Quarantine is a choice.

Count the four possible outcomes

Start with the population

200 defective parts; 9,800 sound parts.

Apply the sensor rates

$200 \times 0.90 = 180$ detected defects.
$9,800 \times 0.05 = 490$ false alarms.

The other two cells. 20 missed defects; 9,310 sound parts without an alarm.

Inspection of 10,000 parts (teaching counts)

180 Defective + alarm	**490** Sound + alarm
20 Defective + no alarm	**9,310** Sound + no alarm

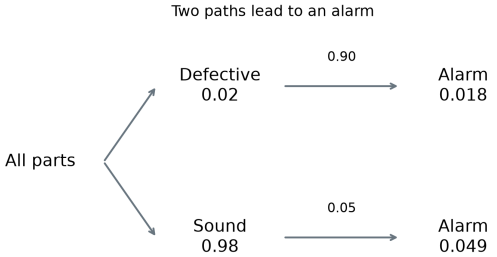
An alarm has two possible explanations

Joint probability

$P(D, A) = 0.02 \times 0.90 = 0.018$

$P(S, A) = 0.98 \times 0.05 = 0.049$

Read the paths. Multiply along a path. Add the paths that end at the same observation.

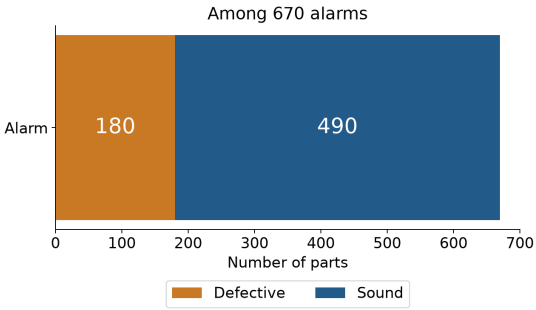


Only 180 of the 670 alarms are defects

$$P(D \mid A) = \frac{180}{180 + 490} = \frac{18}{67} \approx 0.269$$

The denominator has changed

We are now looking only at the alarm group, not at all defective parts.



Bayes' rule keeps the competing explanations

From likelihood to posterior

$$P(Y = k \mid x) = \frac{p(x \mid Y = k)P(Y = k)}{\sum_j p(x \mid Y = j)P(Y = j)}$$

Term	Question it answers
Prior $P(Y = k)$	How common is this class before this observation?
Likelihood $p(x \mid Y = k)$	How compatible is the observation with this class?
Posterior $P(Y = k \mid x)$	How plausible is this class after the observation?

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Read which event is after the conditioning bar

- $P(\text{alarm} \mid \text{defect})$ asks how often defects trigger alarms.
- $P(\text{defect} \mid \text{alarm})$ asks how often alarmed parts are defective.
- These probabilities have different denominators.
- Use Bayes' rule to include the defect prevalence.

Why the largest posterior minimizes errors

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Equal penalties for all mistakes

$$R(k \mid x) = \sum_{j \neq k} P(Y = j \mid x) = 1 - P(Y = k \mid x)$$

For an alarmed part

Predict defective: error probability 0.731.
Predict sound: error probability 0.269.

A restricted objective

This minimizes the number of mistakes, not every possible consequence.

Compute the posterior with three array operations

```
import numpy as np
prior = np.array([0.02, 0.98])  # defective, sound
alarm_rate = np.array([0.90, 0.05])
weights = prior * alarm_rate
posterior = weights / weights.sum()
print(weights)                  # [0.018 0.049]
print(np.round(posterior, 3))   # [0.269 0.731]
assert np.isclose(posterior.sum(), 1.0)
```

Keep the order fixed. Every vector uses [defective, sound].

No alarm does not mean no defect

The remaining group

20 defective parts and 9,310 sound parts have no alarm.

$$P(D \mid \neg A) = \frac{20}{20 + 9,310} \approx 0.00214$$

Small is not zero

About 0.214% of the no-alarm group is defective in this example.

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A different factory can produce a different answer

Keep the sensor unchanged

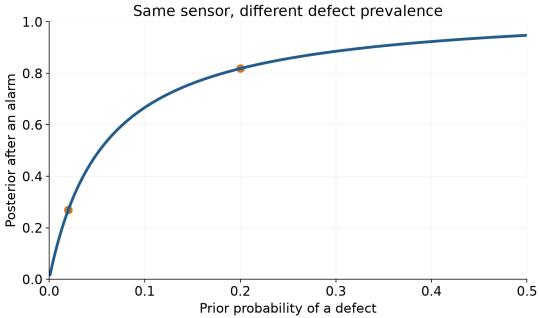
$P(A \mid D) = 0.90$, $P(A \mid S) = 0.05$.

Change the prevalence

2% defective: posterior $\approx 26.9\%$.
20% defective: posterior $\approx 81.8\%$.

Check the deployment population

A training class balance is not automatically the future class prior.



Odds separate evidence from the prior

Posterior odds

$$\frac{P(D \mid A)}{P(S \mid A)} = \frac{P(A \mid D)}{P(A \mid S)} \frac{P(D)}{P(S)}$$

Our alarm

$$18 \times \frac{1}{49} = \frac{18}{49}$$

Meaning. The alarm favors a defect by a factor of 18, but the prior odds favor sound parts by 49 to 1.

More than two classes: normalize all competitors

Class	Prior	Likelihood	Product
A	0.40	0.30	0.12
B	0.30	0.30	0.09
C	0.30	0.1333...	0.04

$$P(Y \mid x) = (0.48,\ 0.36,\ 0.16)$$

Two different tasks. Choose A using the largest product. Divide by 0.25 if probabilities are requested.

Try one calculation without the factory numbers

A new teaching example

$P(C) = 0.10$, $P(+ \mid C) = 0.80$, $P(+ \mid \neg C) = 0.10$.
Find $P(C \mid +)$.

Check your denominator

$$\frac{0.10 \times 0.80}{0.10 \times 0.80 + 0.90 \times 0.10} = \frac{8}{17} \approx 0.471$$

Explain the result

80% is a likelihood, not the requested posterior.

Would you release an alarmed part?

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Return to the factory

After an alarm, $P(D \mid A) = 18/67 \approx 0.269$.

The consequences

Releasing a defective part costs 30 units.
Quarantining a sound part costs 2 units.
Correct actions cost 0 in this simplified model.

A new question

Which action has the smaller expected loss?

Read a loss matrix by action and true state

Chosen action	True defective	True sound
Quarantine	0	2
Release	30	0

Convention used throughout this lecture

Rows are actions. Columns are true states.
$L(a, j)$ is the loss if we choose action a and state j is true.

Read one cell aloud. Release + true defective = a missed defect, loss 30.

Conditional risk is a weighted average of losses

$$R(a \mid x) = \sum_j L(a, j) P(Y = j \mid x)$$

Quarantine

$R(Q \mid A) = 0 \times P(D \mid A) + 2 \times P(S \mid A)$.

Release

$R(R \mid A) = 30 \times P(D \mid A) + 0 \times P(S \mid A)$.

Decision rule. Choose $a^*(x) = \arg\min_a R(a \mid x)$.

Lower probability can still justify quarantine

Action	Exact expected loss	Approximation
Quarantine	$2 \times 49/67$	1.463
Release	$30 \times 18/67$	8.060

Decision

Quarantine has the lower expected loss.

Do not rewrite the probability

The posterior remains 0.269. Only the action rule changed.

Derive the decision threshold from the losses

Let $p = P(D \mid x)$

$$\underbrace{2(1-p)}_{\text{quarantine}} < \underbrace{30p}_{\text{release}}$$

$$p > \frac{2}{30+2} = 0.0625$$

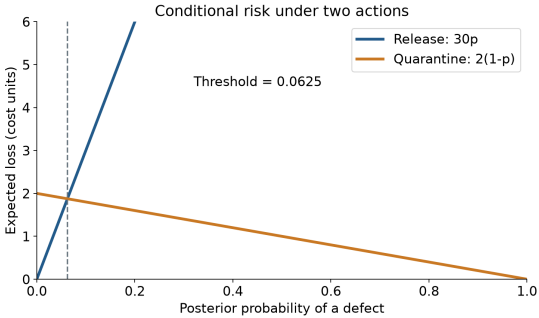
General zero-correct-loss case

$$p > \frac{c_{\text{false alarm}}}{c_{\text{miss}} + c_{\text{false alarm}}}$$

The threshold is where the risk curves cross

Read the graph. Release costs $30p$; quarantine costs $2(1-p)$. The lower curve determines the action.

At an alarm. $p \approx 0.269$ is to the right of the crossing: quarantine.



Changing a cost changes the policy, not the sensor

Miss cost	False-alarm cost	Quarantine threshold
30	2	0.0625
30	10	0.25
90	2	0.0217

Interpretation

Expensive false alarms require stronger evidence. Expensive misses require earlier intervention.

A third action: ask for a review

Action	Defective	Sound
Quarantine	0	2
Release	30	0
Review	0.5	0.5

For the alarmed part

Risks: quarantine 1.463, release 8.060, review 0.5.
Choose review under this stated loss model.

A modelling assumption

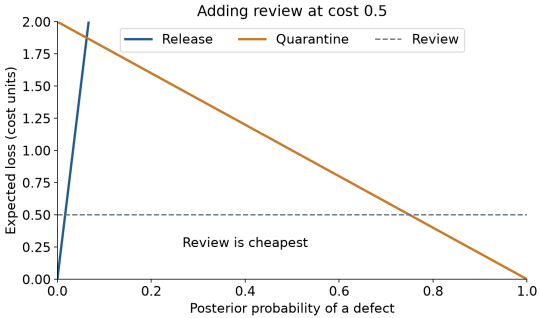
The review cost includes all relevant consequences here; real review can have delays and errors.

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Review occupies the uncertain middle region

Three actions. Release for $p < 1/60$; review for $1/60 < p < 0.75$; quarantine for $p > 0.75$.

At the boundaries. Two actions tie. Pick a consistent tie policy.



The most likely class need not give the best action

Action	True A	True B	True C	Risk
Choose A	0	4	8	2.20
Choose B	2	0	3	1.65
Choose C	5	1	0	3.25

Posterior probabilities

$P(Y \mid x) = (0.60, 0.25, 0.15)$.
Class A is most likely; choosing B has the lowest expected loss.

Turn a loss table into a matrix product

```
post_three = np.array([0.60, 0.25, 0.15])
loss_three = np.array([[0., 4., 8.],
                       [2., 0., 3.],
                       [5., 1., 0.]])
risks_three = loss_three @ post_three
print(risks_three)          # [2.2  1.65 3.25]
print(risks_three.argmin()) # 1: the second action
```

Check the meaning of the index

Python index 1 means the second row, not class label 1 in every dataset.

An everyday example: keep or drop a course?

A separate teaching example

$P(\text{good}) = 0.6$. An engaging class occurs with probability 0.8 in good courses and 0.1 in poor courses.

After one engaging class

$$P(\text{good} \mid \text{engaging}) = \frac{0.48}{0.52} = \frac{12}{13}$$

Losses. Keep a poor course: 10. Drop a good course: 20. Correct choices: 0. Risks are 10/13 versus 240/13; keep it.

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Pointwise decisions minimize overall expected risk

Average over future observations

$$\mathcal{R}(a) = E_X[R(a(X) \mid X)], \qquad \mathcal{R}^* = E_X[\min_a R(a \mid X)]$$

Factory model, with only release or quarantine

Quarantine alarms and release non-alarms. Expected loss per part:

$$\frac{490 \times 2 + 20 \times 30}{10,000} = 0.158$$

Bayes risk. The minimum under the true distribution, specified loss, and available actions. An estimated model need not attain it.

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Can a few words reveal the kind of message?

Two messages, one classification problem

Two messages

One promises a free prize. One arranges a project meeting.

The task

Predict spam or ordinary mail from the words that appear.

A practical question. How can we assign a probability to a whole message?

Message A

FREE prize! Claim your reward.

Message B

Project meeting at 10. Bring your draft.

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Turn words into explicit observations

Message fragment	free present?	meeting present?	free count
free prize	1	0	1
free free reward	1	0	2
project meeting	0	1	0
free meeting room	1	1	1

Choose a representation

Presence and counts answer different questions. Do not switch between them silently.

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The simplifying assumption sits inside each class

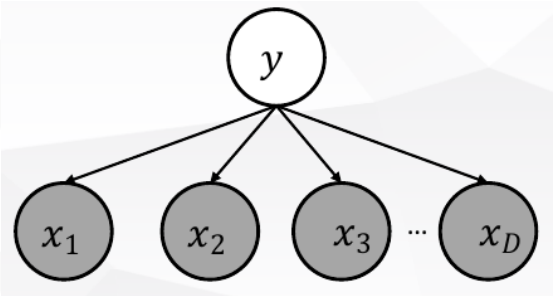
Naive Bayes assumption

$$p(x_1, \ldots, x_d \mid Y = k) = \prod_{j=1}^{d} p(x_j \mid Y = k)$$

Read the graph. Condition on the class at the top. Model each observed feature separately.

Not a statement about language

Real words need not be independent; this is the model assumption.



Why the simplification saves so many parameters

Ten binary features, within one class	Free probabilities
Full joint table: 2^{10} possible patterns	$2^{10} - 1 = 1,023$
Independent Bernoulli features	10

With two classes

2,046 class-conditional parameters become 20, plus one class-prior parameter.

The trade-off

Fewer parameters mean less to estimate, but less freedom to represent interactions.

Conditional independence is not marginal independence

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A small probability model

Equal class priors. In class A, both features have probability 0.9; in class B, both have probability 0.1. They are independent within each class.

$$P(X_1 = 1) = P(X_2 = 1) = 0.5$$
$$P(X_1 = 1, X_2 = 1) = 0.5(0.9^2) + 0.5(0.1^2) = 0.41$$

After mixing classes

$0.41 \neq 0.5 \times 0.5$: the features are not independent overall.

Learn presence probabilities from labelled messages

Training counts	Spam	Ordinary
Number of messages	4	6
Messages containing free	3	1
Messages containing meeting	0	4

Before smoothing

$P(\text{spam}) = 4/10$.
$P(\text{free} = 1 \mid \text{spam}) = 3/4$.

Observation model. One Bernoulli variable per word: present or absent.

An unseen event should not erase every other clue

The zero-frequency problem

No spam message in four training examples contains meeting. A raw estimate gives probability 0.

Add-one smoothing for a binary feature

$$\hat{\phi}_{kj} = \frac{n_{kj} + 1}{n_k + 2}$$

For meeting in spam

$(0 + 1)/(4 + 2) = 1/6$, not 0.

An absent word is evidence in the Bernoulli model

Smoothed probability	Spam	Ordinary
free present	$4/6$	$2/8$
meeting present	$1/6$	$5/8$

Message: free present, meeting absent

$$w_S = 0.4 \times \frac{4}{6} \times \left(1 - \frac{1}{6}\right) \approx 0.2222$$

$$w_O = 0.6 \times \frac{2}{8} \times \left(1 - \frac{5}{8}\right) = 0.05625$$

A complete Bernoulli prediction in code

```
class_count = np.array([4., 6.])
present = np.array([[3., 0.], [1., 4.]])
phi_binary = (present + 1) / (class_count[:, None] + 2)
x_binary = np.array([1., 0.])  # free, meeting
likelihood_binary = (phi_binary ** x_binary *
                     (1 - phi_binary) ** (1 - x_binary)).prod(axis=1)
w_binary = class_count / class_count.sum() * likelihood_binary
print(np.round(w_binary / w_binary.sum(), 3))  # [0.798 0.202]
```

Two dimensions to track. Rows are classes; columns are words. The input describes one new message.

Counts use a different sample space

Training word counts	free	meeting	project	Total
Spam	6	0	2	8
Ordinary	1	5	4	10

Add-one smoothing for a vocabulary of size 3

$$\hat{\phi}_{kj} = \frac{n_{kj} + 1}{\sum_v n_{kv} + 3}$$

New example. These are token counts, not the message-presence counts on the previous slides. Class priors remain 0.4 and 0.6.

Score the message "free free project"

Feature vector

$x = (2, 0, 1)$ in vocabulary order [free, meeting, project].

$$w_S = 0.4 \left(\frac{7}{11}\right)^2 \left(\frac{3}{11}\right) \approx 0.04418$$

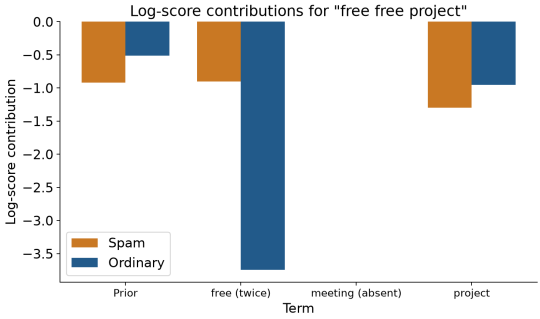
$$w_O = 0.6 \left(\frac{2}{13}\right)^2 \left(\frac{5}{13}\right) \approx 0.005462$$

A common factor cancels. The multinomial coefficient is identical for both classes and is omitted when comparing scores.

Use log scores when the word list gets long

$$s_k = \log P(Y = k) + \sum_j x_j \log \phi_{kj}$$

Read the contributions. Repeated free contributes twice; absent meeting contributes zero. Add log probabilities to avoid tiny products.



Normalize log scores without numerical overflow

```
token_counts = np.array([[6., 0., 2.], [1., 5., 4.]])
phi_count = (token_counts + 1) / (token_counts.sum(axis=1, keepdims=True) + 3)
x_count = np.array([2., 0., 1.])
log_scores = np.log([0.4, 0.6]) + np.log(phi_count) @ x_count
shifted = log_scores - log_scores.max()
post_count = np.exp(shifted) / np.exp(shifted).sum()
print(np.round(post_count, 3))  # [0.89 0.11]
```

Subtracting the same constant is safe

It preserves all score differences and cancels during normalization.

Copying a feature can double-count the evidence

One clue

Prior odds $1:1$ and likelihood ratio $2:1$ give posterior odds $2:1$, or probability $2/3$.

Duplicate the same clue as a new feature

A naive product gives odds $4:1$, or probability $4/5$. No new information was observed.

The problem

Two copies are perfectly dependent within a class. The independence assumption is violated.

A bag of words does not preserve who did what

Sentence A

The dog bites the man.

Sentence B

The man bites the dog.

$$\text{count vector A} = \text{count vector B}$$

An information limit

Any classifier using only this identical count vector must treat the two sentences identically.

Not every discrete feature is a word count

Account feature	Possible values	Distribution
Posting frequency	low, medium, high	Categorical
Profile photo supplied	yes, no	Bernoulli
Number of posts by topic	counts across topics	Multinomial

Categorical smoothing

$$P(X_j = m \mid Y = k) = \frac{n_{kjm} + 1}{n_k + M_j}$$

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The observation determines the distribution family

What is measured?	Possible model	Parameters within each class
Present or absent	Bernoulli	One probability per feature
One of several categories	Categorical	A probability vector
Counts of vocabulary tokens	Multinomial	Word probabilities
Continuous measurement	Gaussian	Mean vector and covariance matrix

Type is only a starting point

A continuous feature is not automatically Gaussian. Check whether the assumption is plausible.

A model family is not yet a fitted model

Object	Example	What is unknown?
Observation X	A binary outcome, 0 or 1	The next outcome
Parameter θ	$P(X = 1) = \theta$	The probability generating outcomes
Data $\mathcal{D}$	Observed past outcomes	Nothing: these are now fixed records

Two fitting ideas

Find the parameter that best explains the data, or update a probability distribution over the parameter.

Ten outcomes are evidence about one probability

A new estimation example

$\mathcal{D} = (1, 1, 0, 1, 0, 1, 1, 1, 0, 1)$.
Seven successes and three failures, independently sampled with a common θ.

Likelihood of this observed sequence

$$L(\theta) = P(\mathcal{D} \mid \theta) = \theta^7 (1 - \theta)^3$$

Keep the interpretation straight. The data are fixed; the parameter is varied.

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The likelihood peaks near the observed fraction

Compare candidate parameters

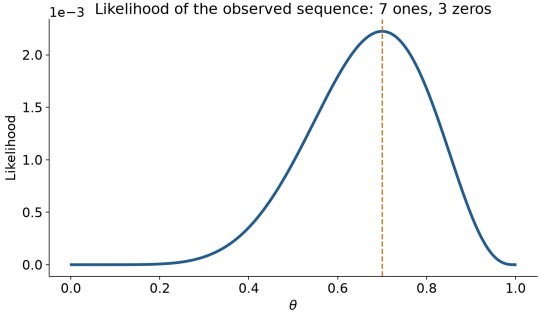
Too small a θ makes seven successes unlikely.
Too large a θ makes three failures unlikely.

The peak

$\hat{\theta} = 7/10 = 0.7$.

Vertical axis

Likelihood of the observed sequence, not a density over the parameter.



Maximum likelihood gives the sample proportion

$$\ell(\theta) = 7 \log \theta + 3 \log(1 - \theta)$$

$$\frac{d\ell}{d\theta} = \frac{7}{\theta} - \frac{3}{1 - \theta} = 0$$

$$7(1 - \theta) = 3\theta \quad \Rightarrow \quad \hat{\theta} = 0.7$$

General result

For s successes in n independent Bernoulli trials, $\hat{\theta} = s/n$.

Class posterior and parameter posterior differ

Earlier classification	Now parameter estimation
$P(Y = k \mid x)$	$p(\theta \mid \mathcal{D})$
Which class produced this observation?	Which parameter values explain the data?
A distribution across class labels	A distribution across possible parameters

$$p(\theta \mid \mathcal{D}) \propto p(\mathcal{D} \mid \theta)\, p(\theta)$$

A prior describes uncertainty before this dataset

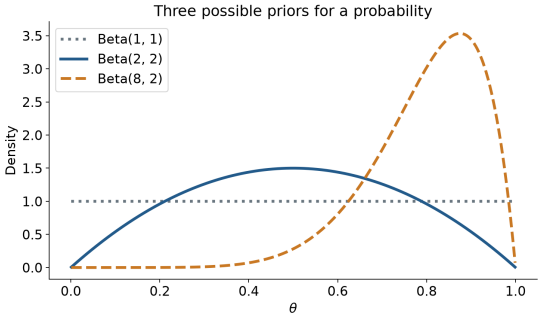
A Beta prior for a probability

$$p(\theta) \propto \theta^{\alpha-1}(1-\theta)^{\beta-1}, \quad 0 < \theta < 1$$

Three examples. Beta(1,1): uniform. Beta(2,2): centered. Beta(8,2): favors larger probabilities.

A choice to justify

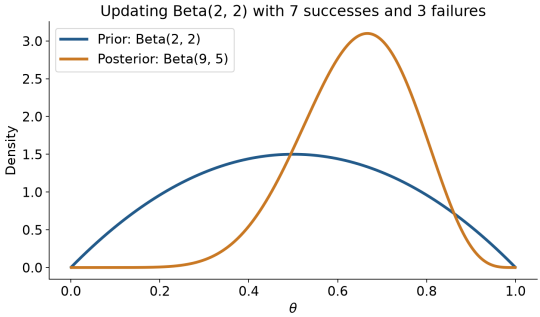
A convenient prior is not automatically a suitable prior.



Updating Beta counts gives a full posterior

$$p(\theta \mid \mathcal{D}) \propto \theta^8(1-\theta)^4 \quad \Rightarrow \quad \theta \mid \mathcal{D} \sim \text{Beta}(9, 5)$$

General update. Beta(α, β) becomes Beta$(\alpha + s, \beta + n - s)$.



Three estimates answer three different questions

Estimate	Formula in this example	Value
Maximum likelihood	$7/10$	0.700
Posterior mean	$9/(9+5)$	0.643
Posterior mode	$(9-1)/(9+5-2)$	0.667

Mode formula condition

$(\alpha - 1)/(\alpha + \beta - 2)$ applies to an interior Beta mode when both parameters exceed 1.

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Prediction averages over parameter uncertainty

$$P(X_{\text{new}} = 1 \mid \mathcal{D}) = \int_0^1 \theta \, p(\theta \mid \mathcal{D}) \, d\theta = \frac{9}{14}$$

One future Bernoulli outcome

Its predictive success probability equals the posterior mean of the parameter.

Do not overgeneralize

For nonlinear predictions, averaging predictions need not equal prediction at the average parameter.

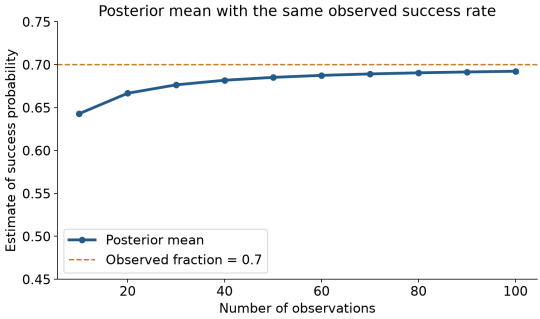
More data reduce the influence of a fixed prior

$$E[\theta \mid \mathcal{D}] = \frac{n}{n+4}\frac{s}{n} + \frac{4}{n+4}\frac{1}{2}$$

Keep the observed success fraction at 0.7

$n = 10$: $9/14 \approx 0.643$.
$n = 100$: $72/104 \approx 0.692$.

Meaning. With a fixed Beta(2,2) prior, the data receive more weight as the sample size grows.



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Updating in batches should give the same answer

Stage	Successes added	Failures added	Posterior
Start	0	0	Beta(2,2)
First batch	3	1	Beta(5,3)
Second batch	4	2	Beta(9,5)

Compare with one batch

All seven successes and three failures at once also give Beta(9,5).

Do not count the first batch twice

The first posterior already contains the first batch.

Check a Bayesian update with a short program

```
alpha_prior, beta_prior = 2., 2.
successes, failures = 7, 3
alpha_post = alpha_prior + successes
beta_post = beta_prior + failures
post_mean = alpha_post / (alpha_post + beta_post)
post_mode = (alpha_post - 1) / (alpha_post + beta_post - 2)
print(round(post_mean, 6), round(post_mode, 6))
# 0.642857 0.666667
assert (2 + 3 + 4, 2 + 1 + 2) == (alpha_post, beta_post)
```

No fitting loop is needed here. The conjugate update has a closed form.

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The same counting idea extends to several outcomes

Three categories

Observed counts $(3, 1, 0)$. Maximum-likelihood probabilities: $(3/4, 1/4, 0)$.

Uniform Dirichlet prior

$$\boldsymbol{\theta} \sim \text{Dirichlet}(1, 1, 1)$$

$$\boldsymbol{\theta} \mid \mathcal{D} \sim \text{Dirichlet}(4, 2, 1)$$

Posterior mean. $(4/7, 2/7, 1/7)$. Add one to each count, then normalize.

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A Gaussian mean can also be learned with a prior

Known observation variance

$X_i \mid \mu \sim \mathcal{N}(\mu, \sigma^2)$, independently; $\mu \sim \mathcal{N}(m_0, v_0)$.

$$v_n = \left(v_0^{-1} + n/\sigma^2\right)^{-1}, \qquad m_n = v_n \left(m_0/v_0 + n\bar{x}/\sigma^2\right)$$

A numerical update

$m_0 = 0$, $v_0 = 4$, $\sigma^2 = 4$, $n = 4$, $\bar{x} = 3$.
Posterior for μ: $\mathcal{N}(2.4, 0.8)$.

A sharper parameter posterior is not noiseless data

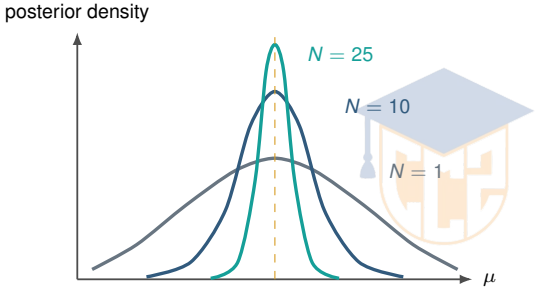
What the original illustration shows

Posterior curves for an unknown mean become more concentrated as observations accumulate.

Two different variances

Uncertainty about the mean can shrink while individual measurements still vary.

Prediction with known noise variance. $\mathrm{Var}(X_{\mathrm{new}} \mid \mathcal{D}) = \sigma^2 + v_n$, not just v_n.



More observations reduce uncertainty about the mean; measurement noise remains.

One last check: all successes with a uniform prior

Three successes, no failures

$\theta \sim \text{Beta}(1, 1)$ before the observations.

Quantity	Answer
Posterior	Beta(4,1)
Maximum-likelihood estimate	1
Posterior mean and next-success probability	4/5
Posterior mode	1, at the boundary

Next. For continuous feature vectors, we need a model for both the centre and the shape.

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Continuous measurements form clouds, not word lists

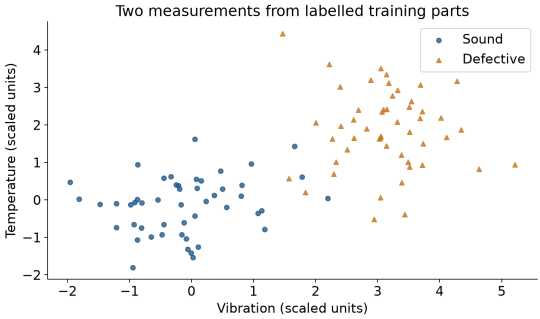
Two measurements per part

$\mathbf{x} = (x_1, x_2)^T$: vibration and temperature in scaled units.

A generative model for each class

$p(\mathbf{x} \mid Y = k) = \mathcal{N}(\mathbf{x}; \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)$.

Question. Which class distribution gives stronger support to a new point?



A two-dimensional density is a surface

Read the three axes

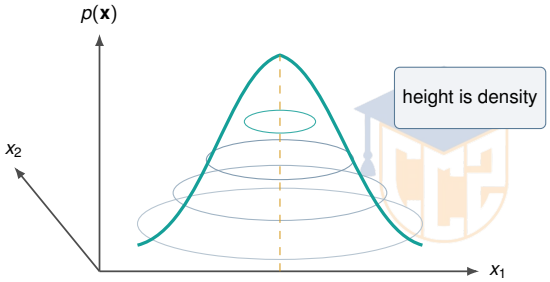
x_1 and x_2 describe the feature location. Height is the density $p(\mathbf{x})$.

A probability is a volume

Integrate the density over a region of the feature plane.

Height is not class probability

A high density means local concentration within this distribution, not certainty about a class.



Probability is the volume under the surface over a region.

A density can exceed one without any contradiction

$$p(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

A narrow distribution

For $X \sim \mathcal{N}(0, 0.01)$, the standard deviation is 0.1 and $p(0) \approx 3.989$.

Probability stays between zero and one

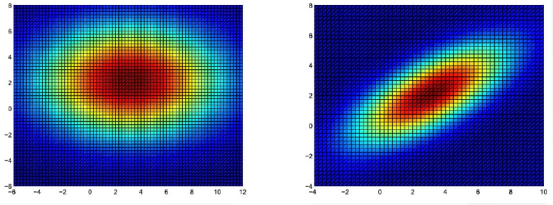
$P(-0.1 \leq X \leq 0.1) \approx 0.683$. It is an area, not the height 3.989.

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The same centre can hide different class shapes

$$\boldsymbol{\mu} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}, \quad \Sigma_{\text{left}} = \begin{bmatrix} 25 & 0 \\ 0 & 9 \end{bmatrix}, \quad \Sigma_{\text{right}} = \begin{bmatrix} 15 & 5 \\ 5 & 5 \end{bmatrix}$$

Compare geometry. The left density is axis-aligned. The positive off-diagonal covariance on the right tilts the contours upward.



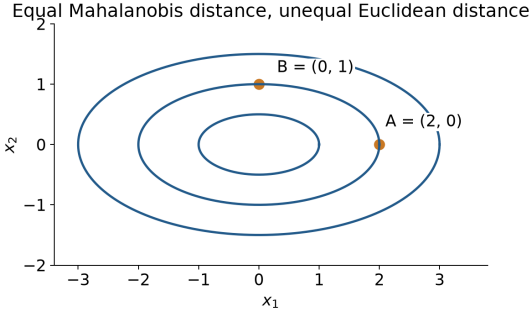
Distance should respect the shape of the cloud

Squared Mahalanobis distance

$$d_M^2 = (\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})$$

With $\mu = 0$ and $\Sigma = \mathrm{diag}(4, 1)$

A=(2, 0) and B=(0, 1) both have $d_M^2 = 1$. Their Euclidean distances are 2 and 1.



Three ingredients determine Gaussian density

$$p(\mathbf{x} \mid k) = \frac{\exp[-\frac{1}{2}(\mathbf{x} - \mu_k)^T \Sigma_k^{-1}(\mathbf{x} - \mu_k)]}{(2\pi)^{d/2}|\Sigma_k|^{1/2}}$$

Ingredient	Role
Mean μ_k	Centre of the class
Inverse covariance Σ_k^{-1}	Shape-aware distance
Determinant $\lvert\Sigma_k\rvert$	Normalizes the spread and height

Assumption for this density formula

The covariance matrix is symmetric and positive definite.

Use a log score to compare Gaussian classes

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Class score, dropping a common constant

$$g_k(\mathbf{x}) = -\frac{1}{2}\log|\Sigma_k| - \frac{1}{2}(\mathbf{x}-\mu_k)^T\Sigma_k^{-1}(\mathbf{x}-\mu_k) + \log\pi_k$$

Three terms. Spread penalty + shape-aware distance penalty + log prior.

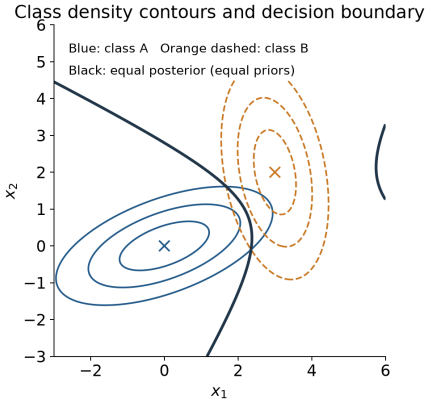
What may be removed

$-d\log(2\pi)/2$ is shared by all classes. A class-specific determinant is not.

Different covariance matrices can bend the boundary

The plotted example. $\mu_A = (0, 0)^T$, $\mu_B = (3, 2)^T$; equal priors. Covariances differ.

Boundary equation. $g_A(\mathbf{x}) = g_B(\mathbf{x})$. The quadratic terms generally do not cancel. The detached black arc is a second branch of the same quadratic boundary.



Shared covariance makes the boundary linear

$$\Sigma_A = \Sigma_B = \Sigma$$

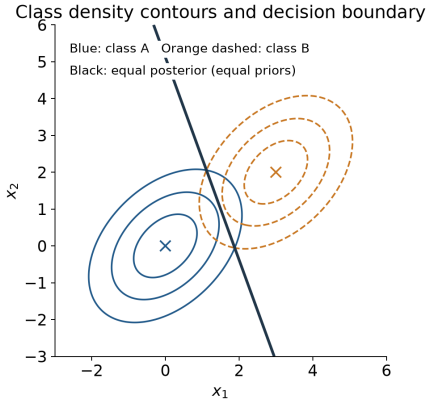
$$g_k(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mu_k - \tfrac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log \pi_k + \text{common terms}$$

After comparing two classes

$$g_A(\mathbf{x}) - g_B(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b$$

Why. The same quadratic term and the same log determinant cancel.

Read contours and the separating line together



In this model. Both classes share covariance with positive correlation and have equal priors.

What remains different. Their centres. The black boundary is straight even though each class has elliptical contours.

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Nearest mean is a special case, not the whole rule

A sufficient set of conditions

Zero-one loss, equal priors, and one shared spherical covariance: $\Sigma_k = \sigma^2 I$.

$$\arg\max_k g_k(\mathbf{x}) = \arg\min_k \|\mathbf{x} - \mu_k\|^2$$

Change the assumption	What changes?
Shared but non-spherical covariance	Use a shape-aware metric
Unequal priors	Add a class preference
Different class covariances	Distance and volume both matter

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Priors move a boundary even when densities stay fixed

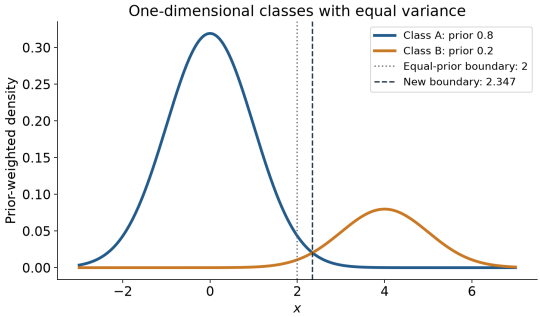
One-dimensional example

$X \mid A \sim \mathcal{N}(0,1)$; $X \mid B \sim \mathcal{N}(4,1)$.

$$x^* = 2 + \frac{1}{4} \log \frac{\pi_A}{\pi_B}$$

Change the priors

Equal priors: $x^* = 2$.
$(\pi_A, \pi_B) = (0.8, 0.2)$: $x^* \approx 2.347$.



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One class can win in both tails

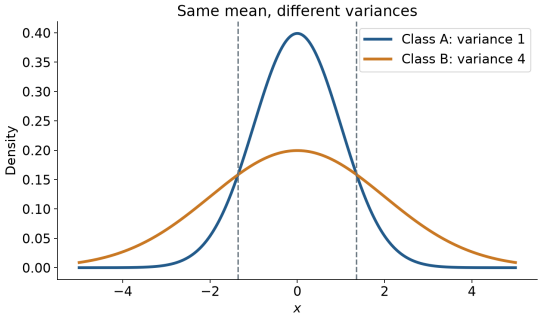
Same mean, unequal variances

$X \mid A \sim \mathcal{N}(0,1)$, $X \mid B \sim \mathcal{N}(0,4)$; equal priors.

$$|x| = \sqrt{\frac{8}{3}\log 2} \approx 1.360$$

Decision regions

Choose A near 0; choose B in both tails.



Diagonal covariance does not guarantee a line

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Gaussian Naive Bayes

Each class has a diagonal covariance matrix; features are independent within that Gaussian class.

Class-specific variances

If the diagonal entries differ between classes, quadratic terms can remain.

Different restrictions

Diagonal within each class $\neq$ shared across classes. Shared covariance is the condition used for a linear boundary.

Inspect every term in a numerical Gaussian score

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Given parameters

$\mu_A = (1,1)^T$, $\mu_B = (5,5)^T$, $\Sigma_A = \Sigma_B = I$, equal priors. New point: $(2,3)^T$.

Term	Class A	Class B
Squared distance	5	13
Log determinant	0	0
Score without common normalizer	$-2.5 + \log 0.5$	$-6.5 + \log 0.5$

$$P(A \mid \mathbf{x}) = \frac{1}{1 + e^{-4}} \approx 0.982$$

Compute a Gaussian score without forming an inverse

```
def gaussian_score(x, mean, covariance, prior):
    delta = x - mean
    sign, logdet = np.linalg.slogdet(covariance)
    np.linalg.cholesky(covariance)  # requires positive definiteness
    if sign <= 0 or prior <= 0:
        raise ValueError("Invalid covariance or prior")
    distance2 = delta @ np.linalg.solve(covariance, delta)
    return -0.5 * (logdet + distance2) + np.log(prior)
```

Input contract. A symmetric positive-definite covariance and a positive prior. The function returns an unnormalized log score.

Start from eight points, not from given parameters

Class A

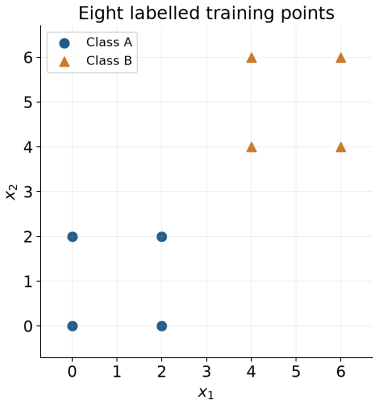
$(0,0), (2,0), (2,2), (0,2)$.

Class B

$(4,4), (6,4), (6,6), (4,6)$.

Our complete task

Estimate the parameters, derive the boundary, then classify new points.



Estimate one mean and one prior per class

$$\hat{\mu}_k = \frac{1}{n_k} \sum_{i:y_i=k} \mathbf{x}_i, \qquad \hat{\pi}_k = \frac{n_k}{n}$$

The eight-point dataset

$\hat{\mu}_A = (1,1)^T$, $\hat{\mu}_B = (5,5)^T$.
$n_A = n_B = 4$, so $\hat{\pi}_A = \hat{\pi}_B = 0.5$.

Keep the labels attached

Compute class means from the samples in that class, not from the entire mixed cloud.

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Centre the samples before measuring their spread

A sample	Centred vector	Product of coordinates
$(0, 0)$	$(-1, -1)$	1
$(2, 0)$	$(1, -1)$	-1
$(2, 2)$	$(1, 1)$	1
$(0, 2)$	$(-1, 1)$	-1

One outer product

$$\begin{bmatrix} 1 \\ -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$$

Maximum-likelihood covariance averages outer products

$$\hat{\Sigma}_k = \frac{1}{n_k} \sum_{i:y_i=k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^T$$

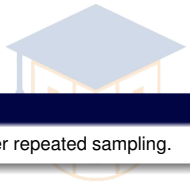
For class A

$$\sum_i (\mathbf{x}_i - \hat{\mu}_A)(\mathbf{x}_i - \hat{\mu}_A)^T = \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$$

$$\hat{\Sigma}_A = \hat{\Sigma}_B = I$$

Divide by n or n minus one? State the estimator

Estimator	Class-A covariance	NumPy setting
Gaussian maximum likelihood	I	`ddof=0`
Unbiased sample covariance	$\frac{4}{3}I$	`ddof=1`



Same numerator, different objectives

Maximum likelihood fits the Gaussian model. The unbiased version corrects the expected covariance estimate under repeated sampling.

Defaults can differ

Use an explicit denominator when matching a derivation to code.

Shared covariance is pooled within classes

$$\hat{\Sigma}_{\text{shared}} = \frac{1}{n} \sum_{k} \sum_{i:y_i=k} (\mathbf{x}_i - \hat{\mu}_k)(\mathbf{x}_i - \hat{\mu}_k)^T$$

Correct within-class result

For these eight points, $\hat{\Sigma}_{\text{shared}} = I$.

Not the covariance around the grand mean

$$\text{Cov}_{\text{all}}(X) = \begin{bmatrix} 5 & 4 \\ 4 & 5 \end{bmatrix} \quad \text{with denominator } n.$$

Fit the parameters directly from the labelled array

```
X = np.array([[0.,0.],[2.,0.],[2.,2.],[0.,2.],
              [4.,4.],[6.,4.],[6.,6.],[4.,6.]])
y = np.array([0,0,0,0,1,1,1,1])
means = np.array([X[y == k].mean(axis=0) for k in (0, 1)])
covariances = np.array([np.cov(X[y == k], rowvar=False, ddof=0)
                        for k in (0, 1)])
priors = np.bincount(y) / len(y)
print(means)  # [[1. 1.] [5. 5.]]
assert np.allclose(covariances, np.array([np.eye(2), np.eye(2)]))
```

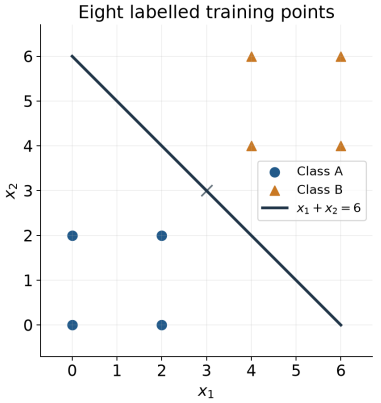
Array convention. Rows are samples; columns are features. Labels are stored separately.

The fitted boundary is x1 plus x2 equals six

$$\|\mathbf{x} - (1,1)^T\|^2 = \|\mathbf{x} - (5,5)^T\|^2$$

$$8x_1 + 8x_2 - 48 = 0 \quad \Rightarrow \quad x_1 + x_2 = 6$$

Regions. Choose A below the line and B above it. Points on the line tie under equal priors and zero-one loss.



Predict a new point with the fitted model

```
new_point = np.array([4., 3.])
new_scores = np.array([gaussian_score(new_point, means[k],
                                      covariances[k], priors[k])
                       for k in (0, 1)])
new_weights = np.exp(new_scores - new_scores.max())
new_posterior = new_weights / new_weights.sum()
print(np.round(new_posterior, 3))  # [0.018 0.982]
print(["A", "B"][new_posterior.argmax()])  # B
```

Geometry agrees. $4 + 3 > 6$, so the point lies in the class B region.

Too few independent directions make covariance singular

Three points on one line

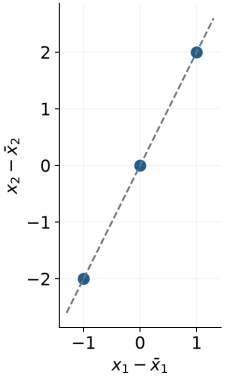
$(0,0), (1,2), (2,4)$ have only one direction of variation.

$$\hat{\Sigma} = \begin{bmatrix} 2/3 & 4/3 \\ 4/3 & 8/3 \end{bmatrix}, \qquad |\hat{\Sigma}| = 0$$

A general rank limit

With n_k samples in d dimensions, rank is at most $\min(d, n_k - 1)$.
Even more samples can be collinear.

Centred observations confined to one direction



Shrinkage gives weak directions some variance

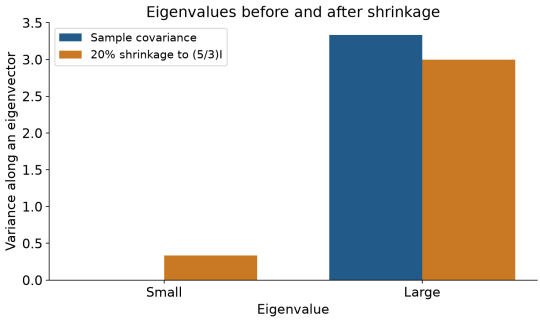
$$\tilde{\Sigma} = (1-\lambda)S + \lambda\tau I, \quad \tau = \mathrm{tr}(S)/d$$

For the collinear example

$\lambda = 0.2$, $\tau = 5/3$. Eigenvalues change from $(0, 10/3)$ to $(1/3, 3)$.

A bias-stability trade-off

The added variance is a modelling choice. Choose its strength using training/validation data, not the final test set.



Covariance structure controls how much must be learned

Structure for two classes, d=20	Covariance parameters
Separate full covariance matrices	$2 \times 210 = 420$
One shared full covariance matrix	210
Separate diagonal covariance matrices	$2 \times 20 = 40$
One shared spherical variance	1

These counts exclude means and priors. A symmetric full matrix has $d(d+1)/2$ free entries.

Choose with evidence

More flexible is not always better when each class has few observations.

Update means and covariance as samples arrive

Keep a count, a mean, and a scatter matrix

$$\delta = \mathbf{x}_{n+1} - \mu_n, \quad \mu_{n+1} = \mu_n + \frac{\delta}{n+1}$$

$$M_{n+1} = M_n + \delta(\mathbf{x}_{n+1} - \mu_{n+1})^T$$

Maximum-likelihood covariance

After n samples, $\hat{\Sigma} = M_n/n$.
For the scalar sequence 2, 4, 6: mean 4, scatter 8, variance $8/3$.

Per-class training. Maintain a separate state for each class; initialize count 0, mean 0, scatter 0.

Test the decision objective, not just accuracy

On 100 labelled test parts	Policy A	Policy B
Missed defects, out of 20	2	1
False alarms, out of 80 sound parts	8	16
Accuracy	90%	83%
Total loss: 30 per miss, 2 per false alarm	76	62

Read both outcomes

Policy A makes fewer errors. Policy B has lower total loss under these stated costs.

Teaching comparison. Invented test counts illustrate the metric trade-off; they are not results from the eight-point dataset.

Check the probability model before accepting its prediction

- Check priors, likelihoods, and posterior normalization separately.
- Use the class covariance estimator stated in the model.
- Do not treat correlated copies as independent evidence.
- Choose the action using its loss, not only the largest posterior.

Finish the factory decision from data to action

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A fresh labelled training sample

10 defective parts: 8 alarms.
90 sound parts: 18 alarms. Use unsmoothed frequency estimates.

For a new alarm

$$P(D \mid A) = \frac{(10/100)(8/10)}{(10/100)(8/10) + (90/100)(18/90)} = \frac{4}{13}$$

Choose an action. Miss cost 30; false-alarm cost 2; review cost 0.5. Risks: release 120/13, quarantine 18/13, review 0.5. Choose review.